Explainability Fidelity and Deployment Cost in a Trustworthy Cervical Cytology Screening Pipeline

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Abstract. "Trustworthy AI" is usually evaluated as if it were one property a pipeline either has or lacks. We show instead that it decomposes into at least three independently-varying claims – which explainability method is most faithful, whether combining two uncertainty signals improves selective prediction, and whether an efficient architecture's efficiency claim holds under real measurement – and that each one resolves differently: cleanly, conditionally, or only under direct measurement. Testing all three side by side on one cervical cytology screening pipeline, rather than picking one and assuming the others follow the same pattern, is itself the point. First, we compare three explainability methods (relevancy propagation, attention rollout, and Grad-CAM adapted for vision transformers) against both behavioral faithfulness (deletion/insertion) and a real, expert-annotated localization ground truth (nucleus boundaries from the SIPaKMeD dataset); relevancy propagation wins cleanly on all three metrics, with no method-ranking flip across conditions, and this ranking replicates exactly on a second, independently-trained backbone (ViT-Small). Second, we test whether composing two independently-useful uncertainty signals – temperature-scaled confidence and conformal prediction-set size – into one abstention rule helps: the answer is conditional, helping exactly where the better individual signal was already struggling (under severe distribution shift) and hurting where it was already excellent (in-distribution and under mild shift). Third, we measure what trustworthiness costs in practice: resource- efficiency claims for lightweight vision transformers, evaluated by parameter count and FLOPs alone, do not reliably predict real CPU inference latency, and returning an explanation alongside a prediction costs roughly 3.3× a bare forward pass on CPU. The pattern across all three – a clean answer, a conditional answer, and an answer that only exists once measured – is the paper's central argument: a single verification strategy (pick the best method once, assume composition helps, trust an architectural efficiency claim) does not work uniformly across a trustworthy-AI pipeline, and each property needs its own, independent test. This is why the three are reported together rather than as three separate papers: the divergence between how they resolve is only visible in a joint test on one fixed pipeline, and would be invisible to three papers each evaluating one property against its own, independently chosen baseline.

Keywords: Explainable AI · Vision transformers · Selective prediction · Conformal prediction · Cervical cytology · Resource efficiency

1 Introduction

Cervical cancer screening in specialist-scarce settings is a plausible target for lightweight, on-device AI assistance, but a screening tool that fails silently – misclassifying confidently under conditions its training data didn't cover – is more dangerous than one that fails and says so. Prior work established that a lightweight hybrid vision transformer (EfficientViT-M0) matches or exceeds much larger CNN and full-size ViT baselines on in-distribution cervical cytology accuracy, and that this same backbone can collapse to near-chance accuracy under cross-dataset distribution shift with no static architectural property predicting which models fail safely. A companion study asked whether any of four uncertainty-quantification methods' native confidence signals correctly flag that collapse, and found a genuinely mixed answer: MC Dropout's entropy signal is the only one that substantially improves shift-failure detection on the severe shift condition, while a deep ensemble – the best method in-distribution and under mild shift – does not transfer that advantage to the severe case.

These two prior findings already show that no single fixed choice (backbone, uncertainty method) is safe across all conditions – but each paper tested only one property of the pipeline. The question this paper asks is broader: if trustworthiness is not one property but several, do they all fail to compose the same way, or does each one behave differently under scrutiny? We answer this by testing three additional properties of the same pipeline side by side, deliberately choosing properties that a reader might otherwise assume behave alike: **explainability method choice** (does picking the best explainability method matter as much, and in the same condition-dependent way, as picking the best uncertainty method?), **uncertainty-signal composition** (given that no single uncertainty method is reliably best, does *combining* two different native signals into one system help, or does composition just average away whichever signal was better?), and **measured deployment cost** (does a lightweight-architecture claim, evaluated only by parameter count and FLOPs, actually hold once CPU latency and the cost of an explanation are measured directly?). The answers turn out to differ in kind, not just in degree: explainability method choice resolves cleanly to one winner across all metrics tested; signal composition resolves conditionally, flipping between helpful and harmful depending on shift severity; and the deployment-cost claim resolves only once measured, contradicting what parameter count and FLOPs alone would predict. This three-way divergence – not any single result – is this paper's contribution: it demonstrates that a trustworthy-AI pipeline cannot be verified with one strategy applied uniformly to every component.

Our contributions are threefold, one per property tested. First, a three-way explainability comparison (relevancy propagation vs. attention rollout vs. Grad-CAM) on deletion/insertion faithfulness *and* real expert-annotated nucleus-localization ground truth, where relevancy propagation wins unambiguously on all three metrics – a cleaner result than either companion study's uncertainty-method comparisons produced, and evidence that this property, at least, does not need the condition-by-condition re-testing the other two do. Second, the first system-

level test of whether combining two uncertainty signals for selective prediction helps: a conditional answer (helps under severe shift, hurts in-distribution and under mild shift) that extends the "no single best method" finding from individual methods to method combinations, and shows that composition is not a safe default even when each signal alone is well understood. Third, a measured, not estimated, resource-efficiency and deployment-cost analysis: parameter count and FLOPs do not reliably predict CPU latency across a 12-model registry, and an explanation costs $\sim 3.3\times$ a bare forward pass – a reminder that even the most "static" of the three properties tested here still required direct measurement, not architectural reasoning, to characterize correctly.

2 Related Work

2.1 Transformer explainability methods

Attention rollout [2] treats a transformer's own attention weights, chained across layers via a fixed recurrence, as a relevance signal, on the premise that attention indicates where a model "looks." This premise has itself been contested in the broader literature examining whether attention weights are faithful explanations of a model's actual decision process rather than merely correlated with it – a concern this paper's own faithfulness metrics (Section 3.3) are designed to test empirically rather than assume either way. Relevancy propagation [1] extends attention rollout by weighting each transformer block's attention matrix with its gradient with respect to the target class before the same rollout aggregation, making the resulting relevance map class-discriminative rather than a fixed structural property of attention, and is the primary method evaluated here specifically because it addresses the class-agnosticity critique of raw attention directly. Grad-CAM [3] was originally designed for CNN feature maps, where a single late convolutional layer naturally encodes class-relevant spatial structure; its adaptation to transformers by targeting a block's token-wise activations instead of a convolutional feature map is common practice in the literature, but the specific failure mode we identify and document (Section 3.2) for CLS-token-pooled ViTs – an all-zero heatmap if the wrong block is targeted – does not appear to be widely flagged in prior adaptations, despite being silent rather than an error.

2.2 Selective prediction and conformal calibration

Split conformal prediction provides a distribution-free, finite-sample coverage guarantee from a single calibration split, at the cost of producing a set of plausible labels rather than one score; its prediction set size is a natural, model-agnostic uncertainty proxy, but one that is coarse (a small integer count) compared to a continuous confidence score. MC Dropout [4], deep ensembles [5], and evidential deep learning [6] are the three uncertainty-quantification methods our companion study compares against temperature-scaled softmax; this paper's system-integration question – whether combining temperature-scaled confidence with

conformal set size outperforms either alone – is, to our knowledge, not directly addressed by prior work proposing either signal independently, since each is typically evaluated as a self-contained method rather than as one input to a combined abstention rule.

2.3 Faithfulness evaluation

Deletion/insertion faithfulness [7] evaluates an explanation by its behavioral effect on the model itself (does removing or revealing the patches an explanation calls important actually change the model's prediction), rather than by agreement with a human-annotated region, and is the behavioral-faithfulness metric used throughout this paper. Evaluating localization against SIPaKMeD's own expert-marked nucleus polygons, rather than a segmentation model's derived output, avoids compounding this paper's explainability evaluation with a second model's own errors.

2.4 Closely related prior work

Directly adjacent to this paper's own scope: a stacking-ensemble approach combined with explainable AI for cervical cancer diagnosis [12], which this paper differs from by evaluating explainability methods against a real expert-annotated localization ground truth and behavioral faithfulness metrics rather than an ensemble-accuracy framing; and a lightweight-ViT-plus-explainability design for plant disease diagnosis, "MaizeFormerX" [13], which motivates this paper's own choice to hold efficiency and explainability as joint, not competing, design goals. Two further papers sit in adjacent clinical-AI territory: a semi-supervised retinal disease recognition system [14] and an explainable multimodal AI framework with reinforcement learning for post-surgical clinical decision support [15], both part of a broader body of work in trustworthy, explainable clinical AI of which this paper's cervical cytology system is one instance.

3 Methods

3.1 Backbone

The winning in-distribution backbone from our prior efficiency benchmark, EfficientViT-M0, uses a hybrid convolutional + cascaded-group-attention design. Relevancy propagation and attention rollout are both defined for standard multi-head self-attention transformer blocks; EfficientViT-M0 has no established propagation rule for either method. Rather than force an ad hoc extension onto an architecture these methods were not designed for, this paper uses **DeiT-Tiny** [9] – a standard `timm VisionTransformer` achieving 0.9606 balanced accuracy on the same SIPaKMeD in-distribution task in our prior benchmark (vs. EfficientViT-M0's 0.9654) – retrained fresh for this paper.

3.2 Explainability methods

All three methods produce a relevance map over the model's 14×14 patch grid (DeiT-Tiny, 224px input, 16px patches), upsampled to image resolution. **Attention rollout**: raw softmax attention, averaged over heads, chained across all 12 blocks via the standard rollout recurrence. Class-agnostic. **Relevancy propagation**: identical rollout recurrence, but each block's contribution is $\mathrm{ReLU}(\nabla_{\mathrm{attn}} \odot \mathrm{attn})$, averaged over heads, where the gradient is with respect to the model's own predicted class's logit. Class-discriminative. **Grad-CAM (ViT-adapted)**: gradient-weighted channel importance applied to a transformer block's token activations rather than a CNN's convolutional feature map. *Implementation note*: the target layer must be the *second-to-last* transformer block, not the last. DeiT-Tiny pools only the CLS token for classification, so no layer downstream of the final block mixes information back into patch-token positions – verified directly by inspecting gradients: every patch token's gradient at the last block's output was exactly zero, while the second-to-last block's patch-token gradients vary normally.

3.3 Faithfulness and localization metrics

Deletion/insertion AUC [7]: patches are ranked by a method's own relevance score and progressively replaced with (deletion) or revealed from (insertion) a heavily Gaussian-blurred version of the image, tracking the model's predicted-class probability at each step; the area under the resulting probability-vs-fraction curve is reported (lower is better for deletion, higher is better for insertion). **Nucleus concentration ratio**: SIPaKMeD ships real, expert-marked nucleus boundary polygons per cell; we persist these in crop-local pixel coordinates and compute the fraction of a heatmap's total relevance mass falling inside the nucleus region, divided by the nucleus's share of image area. A ratio of 1.0 is exactly chance level; higher values mean a heatmap concentrates relevance in the nucleus more than chance would predict – clinically the correct place to concentrate.

3.4 Calibration, conformal prediction, and the combined selective-prediction signal

Temperature scaling [8] is fit on a held-out calibration split disjoint from both training and test data. Conformal prediction sets are built via a resampling-averaged split-conformal threshold. The combined selective-prediction signal rank-normalizes temperature-scaled max-softmax confidence and (inverted) conformal prediction-set size independently, then averages their ranks.

3.5 Efficiency and latency measurement

Parameters and FLOPs (via `torch.utils.flop_counter.FlopCounterMode`) are architecture properties independent of trained weight values, so measured without any training. CPU latency and memory (isolated per model in its own

subprocess) are measured with untrained weights for the same reason. The "cost of an explanation" is measured as a full forward+backward+rollout pass for relevancy propagation's own predicted-class explanation, against a bare forward pass, both on CPU.

4 Experiments

4.1 Explainability results

Table 1. Explainability method comparison, 50 SIPaKMeD test images, all matched to a real nucleus polygon.

Method	Deletion AUC↓	Insertion AUC↑	Nucleus conc.↑ (1.0 = chance)
Relevancy propagation	**0.734**	**0.970**	**2.714**
Grad-CAM (ViT-adapted)	0.818	0.877	1.472
Attention rollout	0.825	0.936	1.177

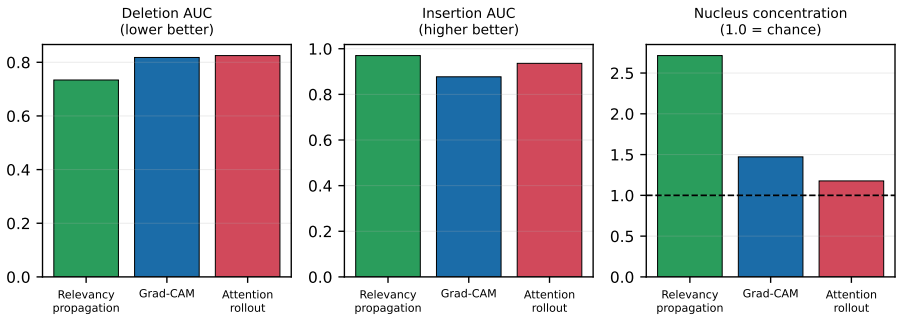


Fig. 1. Deletion AUC, insertion AUC, and nucleus concentration ratio for the three explainability methods. Relevancy propagation wins on all three, with the widest margin on nucleus concentration.

Relevancy propagation wins every comparison, and by a wide margin on nucleus concentration specifically (2.71× chance vs. attention rollout's 1.18× – barely above chance). We attribute this to a mechanistic difference rather than measurement noise: attention rollout is class-agnostic, an inherent disadvantage on an inherently class-discriminative faithfulness test, while relevancy propagation is both class-discriminative *and* integrates the gradient-weighted signal across all twelve transformer blocks.

4.2 A second backbone: does the ranking generalize?

Relevancy propagation's clean, unconditional win is the one result in this paper that looked, on its face, like it might not need condition-by-condition re-testing (Section 2, contribution 1) – but that claim rests on a single backbone (DeiT-Tiny), so we re-ran the identical comparison on ViT-Small, a structurally identical `VisionTransformer` (same `global_pool="token"`, `num_prefix_tokens=1`, 12 blocks – verified directly, not assumed) trained fresh with the same recipe and seed.

Table 2. Explainability ranking on a second backbone (ViT-Small) vs. the original (DeiT-Tiny).

	DeiT-Tiny (original)			ViT-Small (2nd backbone)		
Method	Del. AUC↓	Ins. AUC↑	Conc.↑	Del. AUC↓	Ins. AUC↑	Conc.↑
Relevancy prop.	**0.734**	**0.970**	**2.71**	**0.816**	**0.965**	**4.20**
Grad-CAM	0.818	0.877	1.47	0.888	0.895	1.35
Attn. rollout	0.825	0.936	1.18	0.888	0.927	0.97

The ranking replicates exactly: relevancy propagation wins all three metrics on ViT-Small too, with an even wider nucleus-concentration margin (4.20× chance vs. its closest competitor's 1.35×) than on DeiT-Tiny. This is the one property in this paper's three-way comparison that generalizes cleanly across a backbone change without qualification – in contrast to Section 4.2's uncertainty-composition result, which flips direction depending on shift severity, and Section 4.4's efficiency claims, which required direct measurement precisely because they did *not* generalize from architectural reasoning. The generalization claim is scoped deliberately: both backbones are pure multi-head self-attention `VisionTransformer` instances of different widths and depths, not different architecture families – whether the ranking survives a hybrid conv+attention backbone remains untested, since relevancy propagation is not defined for EfficientViT-M0's architecture (Section 5, External validity).

4.3 System-integration results

Two proposed sub-claims about safe composition hold by construction given this project's post-hoc-attachment architecture, and were verified rather than assumed: temperature scaling changed **zero** argmax predictions on the calibration split, and explainability computation is a read-only analysis that cannot touch model weights. The moderate- and severe-shift conditions below reuse the Herlev [10] and Mendeley LBC [11] datasets from our companion cross-dataset shift study.

Composition is not a uniform win. In distribution and under moderate shift, confidence alone is already the best individual signal, and combining it with the

Table 3. Selective-prediction risk-coverage AUC (lower is better): confidence alone vs. conformal set size alone vs. combined.

Condition	Confidence only	Set size only	Combined
In-distribution	**0.0050**	0.0834	0.0173
Herlev (moderate shift)	**0.1190**	0.3281	0.1793
Mendeley LBC (severe shift)	0.3244	0.4395	**0.2855**

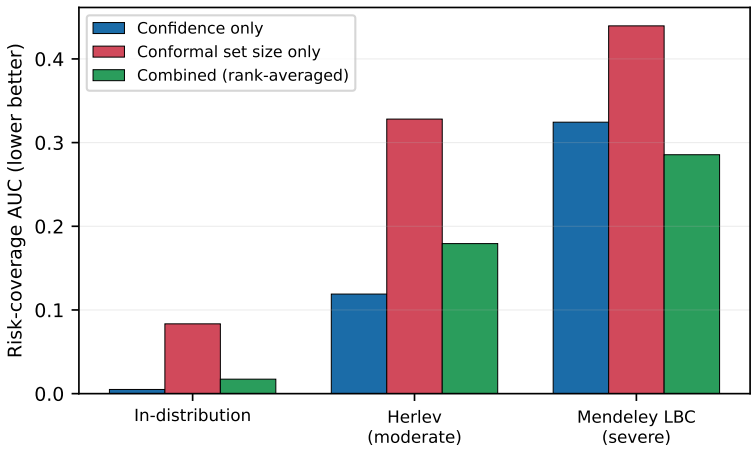


Fig. 2. Risk-coverage AUC for confidence alone, conformal set size alone, and the rank-averaged combination, at three conditions. Combining hurts in-distribution and under moderate shift but helps under severe shift.

noisier discrete conformal-set-size signal dilutes it. Under severe shift, however, the combined signal is the best of the three, beating confidence alone. This mirrors the shape of our companion uncertainty-method finding (no single choice wins everywhere), now extended to combinations of signals.

4.4 Resource efficiency and deployment robustness

Across a 12-model registry, EfficientNet-Lite0 (3.38M params, 0.77G FLOPs) has more than 4× ResNet-50's CPU latency (137.7ms vs. 32.0ms) despite ResNet-50 having roughly 10× the FLOPs. Even EfficientViT-M0, the overall winner on params+FLOPs+accuracy, is not the fastest model in wall-clock CPU terms: DeiT-Tiny (15.4ms) and LeViT-128 (17.5ms) are both faster despite 10–14× more FLOPs. EfficientViT-M0, evaluated under synthetic JPEG compression, Gaussian blur, and brightness/contrast shift, degrades gracefully under all three, but Gaussian blur is by far the most damaging – costing 32.8 points of balanced accuracy at its most severe tested level, roughly double JPEG's worst case (15.5 points) or brightness/contrast's (18.1 points). A synthetic OOD sanity check shows mean confidence dropping from 0.985 (real in-distribution images) to 0.743 (OOD) – real and meaningful, but far above the 0.20 chance level for this 5-class task. On CPU, a bare DeiT-Tiny forward pass has a median latency of 49.6ms; a prediction that also returns a relevancy-propagation explanation costs 164.1ms – about 3.3× the bare-prediction cost.

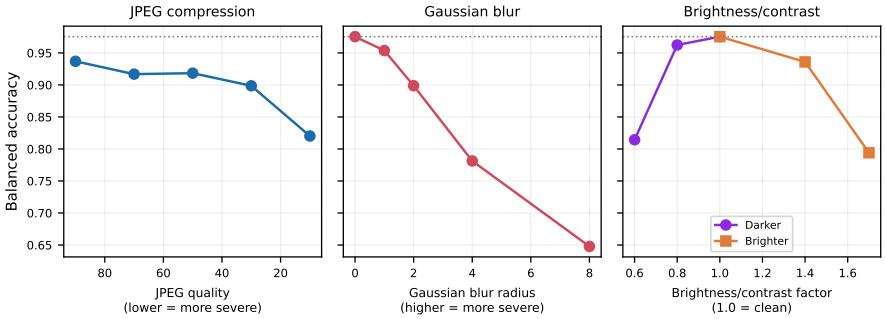


Fig. 3. Balanced accuracy vs. corruption severity for three synthetic degradations (EfficientViT-M0, SIPaKMeD test set). The dotted line marks the clean baseline (0.975). Gaussian blur causes the steepest decline.

5 Discussion

Two threads run through this study. First, explainability method choice produced the cleanest, most unambiguous result so far – relevancy propagation

wins every metric, with no condition-dependent flip – while uncertainty method choice and uncertainty-signal composition both produced conditional, no-single-winner results. This asymmetry suggests that for this task and backbone family, picking the right explainability method is a more reliably beneficial engineering decision than picking (or combining) the right uncertainty-quantification method, where the honest answer is "it depends on the operating condition." Second, every efficiency and robustness claim required direct measurement to be trusted: FLOP count did not predict CPU latency; and OOD confidence dropped only partially, not to the clean near-chance level a reader might assume from an "OOD detection" framing.

6 Threats to Validity

Internal validity. The explainability comparison (Table 1) uses 50 test images, 10 per class, from one fixed stratified sample; a different sample could shift the exact deletion/insertion/nucleus- concentration numbers, though the margin by which relevancy propagation wins on all three metrics is large enough that a resampling-driven reversal of the ranking is unlikely. The OOD sanity check's 25-image sample is smaller still, and its qualitative conclusion (confidence drops but not to chance) should be treated as suggestive rather than a precise estimate for this reason. CPU latency figures (Section 4.4) were measured on one shared consumer laptop rather than isolated benchmark hardware; a ∼30% run-to-run swing was observed for the same model, so the exact 3.3× explanation-cost multiplier should be read as approximate, not a precise constant.

External validity. The explainability ranking (Section 4.1) was re-tested on a second backbone, ViT-Small, and replicated exactly (Section 4.1.1) – but both DeiT-Tiny and ViT-Small are standard multi-head self-attention transformers of the same architectural family, chosen specifically because relevancy propagation is defined for them and not for EfficientViT-M0's hybrid design; this closes the single-backbone gap only within that family. The selective-prediction composition result and the degradation/OOD findings were not re-tested on ViT-Small and still use DeiT-Tiny alone. Whether any of these three results transfer to a hybrid conv+attention backbone, or to a different medical imaging task entirely, remains untested. Single-institution data (SIPaKMeD, ∼4,000 images) throughout.

Construct validity. Deletion/insertion AUC and nucleus concentration ratio are both well-established or directly interpretable metrics (Section 3.3), reducing the risk that the explainability ranking is an artifact of an unusual scoring choice; using SIPaKMeD's own expert-marked nucleus polygons as ground truth, rather than a segmentation model's derived output, specifically avoids compounding this paper's evaluation with a second model's own errors. No clinical or expert-review validation of explanation localization fidelity was performed – agreement with a nucleus boundary is not equivalent to a cytotechnologist judging an explanation clinically meaningful, a distinction stated explicitly rather than implied away.

7 Limitations

Backbone mismatch: the explainability and system-integration results use DeiT-Tiny, not the efficiency benchmark's overall winner, EfficientViT-M0, since relevancy propagation is not defined for the latter; the explainability ranking specifically was confirmed on one additional same-family backbone (ViT-Small, Section 4.1.1), but whether any of this paper's results transfer to a hybrid conv+attention backbone remains untested. Single-institution, small-scale data (SIPaKMeD, ∼4,000 images). CPU latency measured on one shared consumer laptop, not dedicated hardware; a ∼30% run-to-run swing was observed for the same model, though relative ordering across models was stable. OOD sanity check uses a small, synthetic sample (25 images) rather than a natural out-of-domain dataset. No clinical or expert-review validation of explanation localization fidelity.

8 Conclusion

Three properties of one trustworthy-AI pipeline, tested side by side, do not resolve alike. Relevancy propagation is a clearly better explainability method than attention rollout or Grad-CAM for this cervical cytology classifier, on both behavioral faithfulness and real expert-annotated localization ground truth – a clean, condition- independent result that we further confirmed replicates on a second, independently-trained backbone rather than resting on one lucky model. Composing two uncertainty signals for selective prediction, by contrast, helps only under the specific condition where the better individual signal was already failing – a conditional result that would have been reported backwards by an evaluation run under only one shift severity. Resource-efficiency and robustness claims for lightweight vision transformers resolve only under direct measurement, in both directions: FLOP-efficient does not mean latency-efficient, and "confidence drops on OOD input" does not mean "confidence reaches a safe abstention level." No single verification strategy – pick the best method once, assume composition helps, trust an architectural efficiency claim – would have correctly characterized all three properties of this one pipeline; each required its own test, on its own terms.

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